

SCHEDULE -I

SPECIFICATIONS FOR CONSTRUCTION OF CONVENTIONAL TYPE 1800 MT GODOWN

1. Godown Centre to Centre 41.85m x 21.80m
2. Out side to Out side = 42.31m x 22.26m
3. Road side Varandah Width: 1.83m min. on one side of the godown and other side of godown Isolated platform of 0.9 m wide to be provided in front of shutter only (size 2.44 m X 0.9 m).
4. Plinth Height : For road-fed - 0.80 m
5. No. of Stacks in each Compartment 12 Nos.
6. Size of Stacks = 6.10m x 9.15m (20' x 30')
7. Deleted
8. Godown height at eve in respect of road-fed =5.60m.
9. Varandah Truss height on road side = 3.35m above plinth level.
10. No. of rolling shutters = 06 Nos. of size 1.83 m X 2.44 m each
11. Top Ventilators size 1.50m x 0.60 m = 18 Nos. in both long walls
12. V8 ventilators 0.39 m X 0.80 m (Bottom inside air entrant) and Expanded metal on outer walls 0.6 m X 0.6 m (Bottom out side air entrant)
13. R.C. Column to be provided at 4.65 c.c. in long wall & 4.36 m c/c in GW & P.W.
14. Roof to be provided with ACC/CGI/ pre coated galvanized iron profile sheets per CPWD specification 2009, volume-1 roofing materials over tubular roof trusses.

Any specifications not mentioned above will be as per the MPSWC existing specifications for conventional godowns. In case MPWLC does not have any specifications, CWC specifications for godowns to be followed.

In case specifications not covered by MPWLC and CWC then relevant IS code for construction of godowns to be followed.

Tubular Trusses: As per the wind zone classified in the relevant latest BIS code in the normal wind zone: 200 Kg/Sq.m and 150 Kg/Sq.m. or as per actual requirements.

A. ANCILLARIES REQUIRED

Office Block, Sentry post, Weigh Bridge with cabin, Compound wall, Electrification, Water supply.

B. DETAILS OF GENERAL SPECIFICATION FOR GODOWN:

1. **FOUNDATION:** The depth of foundation is proposed for Gravelly Ordinary Soils at 1.30 M for columns and 1.20 M for panel walls (foundation design is based on for soil bearing capacity of 10.93 tonnes per sq.mt). For other type of soils, foundation should be suitably designed. Cement concrete (1:5:10) (1 cement: 5 coarse sand: 10 stone aggregate of 40 mm nominal size) is to be provided under columns and panel wall respectively. R.R. Masonry/Brick Masonry in cement mortar (1:6) (1 cement : 6 coarse sand) is to be provided for foundation and super-structure. The good quality of earth / sand shall be provided under floor.
2. **PLINTH BEAM, TIE BEAM & COLUMNS:** R.C.C. mix of 1:1½:3 (1 cement: 1½ coarse sand: 3 stone aggregate of 20 mm nominal size) is to be provided in column, foundation of column, plinth beam, lintel beam around godown at 2.44m level and tie beam at roof level in long walls.
3. **SUPER STRUCTURE:** All the walls are to be provided with 34 cm thick in brick masonry or 38 cm thick in RR masonry with cement mortar (1:6) (1 cement: 6 coarse sand).
4. **SUN SHED / CHAJJA:** To be provided as per requirements.
5. **FINISHINGS:** 12mm thick cement plaster with cement mortar (1:6) (1 cement: 6 fine sand) on both sides i.e. internal & external walls, 6mm thick plaster with cement mortar (1:3) on exposed surface of columns with 3 coats of white wash on inner walls & colour wash/ snowcem on external walls.
6. **FLOORING:** (A) 230 mm thick sand filling for base. (B) 100 mm thick P.C.C. (1:4:8) (1 cement: 4 coarse sand: 8 coarse aggregate 40 mm

nominal size) (C) Fixing of glass strips 50 x 4 mm size in flooring for panels. 50 mm thick C.C (1:2:4) (1 cement: 2 coarse sand: 4 coarse aggregate 20 mm nominal size) flooring with neat cement punning.

7. **ROOFING:** Tubular trusses on RCC columns with purlins for fixing ACC/CGI. Pre-coated galvanized iron profile sheet with zinc coated sheeting. as per CPWD Specification, 2009, Vol.1 In each compartment 8 Nos. of fiber glass (Translucent) sheets are to be provided (four numbers on each slope).

- 7.2 Galvalume sheet roofing can be provided as an alternative with following specifications :-

7.2.1 The roof of the structure shall be of single span structural with self supporting / pre-engineered structure without any truss or what so ever, at a height not less than 5600 mm from the plinth level to the tie level at the column ends, both for rail-fed and road-fed storage structures. The roofing system should be designed to withstand the wind load (as per IS part 3) & earthquake load (as per IS:893). The design should also comply with any other relevant IS codes. The steel sheet shall be 914 mm wide (Tolerance +/- 2 mm) having minimum thickness 1.03 mm (Tolerance +/- 0.02 mm). The minimum yield strength of the steel sheet should be 350 MPa. The steel sheet should be alloy coated on both sides as per following standards:-

- a) The alloy coating on the steel sheet should be of 55% aluminium, 43.5% zinc and 1.5% Silicon. The minimum thickness of coating shall be 20 microns on outer surface and 10 microns on the inner surface. The thickness of the coated steel should not be less than 1.03 mm including BMT, alloy coating and priming/paint coating.

OR

- b) The steel sheet should have zinc coating of minimum 120 GMS, total of both surfaces as per IS : 277. The minimum total coated thickness of sheet shall be 0.50 mm.

- 7.2.2 Transparent/translucent sheets of about 2% of total area of the roof and evenly distributed may be provided for natural light with sheet made of

polycarbonate material (Poly layer) of minimum thickness 2 mm. The polycarbonate sheets of size 600 mm x 1200 mm having thickness of 2 mm may be installed in the roof at a distance of 900 cm along the length of the building for natural light. The fixing of polycarbonate sheet shall be water tight and structurally sound. M.S. wind ties of size 40 x 6 mm may be provided on translucent sheets for structural safety/stability.

7.2.3 In a godowns of 1800 MT capacity 9 Turbo Ventilators (Aluminium) of size 24", capacity 2200 CFM are required in each compartment.

7.2.4 The design of the roofing shall be in accordance with the general constructional practices and relevant Indian Standard codes for loading standards (IS : 875).

8. WINDOWS, VENTILATORS, ROLLING SHUTTERS:

Rolling shutters (clear opening) of size 1.83m x 2.44m fixed on wall, Air entrant (Windows) of size 0.62m X 0.62 m and top ventilators of size 1.50m X 0.60m with angle iron.

C. ROADS: 27.5 cm thick W.B.M road (Metal Road) having clear width of 6.7m if there is one row of godowns and 15 m wide road in between two rows of godowns with necessary layers of gravel base, metal layer to withstand the Traffic, is to be laid with 4 cm thick premix carpet. Thickness of WBM may be more in case of poor soil having less CBR value.

D. COMPOUND WALL: 1.98 m high boundary wall in brick work (1:6) or RR. Masonry above Ground level with 0.6 m High, 7 rows of barbed wire / concertina coils with angle iron posts 2.44 m C/C on top of wall is to be provided. 15 mm thick plastering in cement mortar (1:6) on rough side of the wall and 12 mm thick plastering with cement (1:6) on plain side of the wall as of brick masonry and pointing ruling/raised & cut on either side of the wall for RR masonry shall be provided.

E. GATE AT MAIN ENTRANCE: 6.10m wide steel gate with tubes and 0.91m wide wicket gate with tubes / angle iron.

F. ELECTRICAL INSTALLATION: External tube light / sodium vapour fittings on out side of godowns at equal intervals and providing of necessary 3 pin

power / light plug points near rolling shutters. Suitable lighting to be provided inside the godowns as well as on platform with the sodium vapour or fittings with CFL lamp lighting fixtures for operation of godowns during late hours.

- G. WATER SUPPLY:** Suitable arrangement for purifying water to overhead tank for supply to office block and drinking water provision for labour / Staff / Officers and others to be provided.
- H. External Drainage :-** The campus shall be provided with well designed drainage as per site conditions .
- I. SEWERAGE :-** Sewage shall be treated as per Pollution Control Board norms prevailing in area .

OFFICE BUILDING : Plinth area 78 Sq.m. including toilets

OFFICE BUILDING SPECIFICATION :

- 1. FOUNDATIONS:** Adequate Depth of foundations shall be provided with leveling course of CC (1:5:10) (1 cement: 5 coarse sand: 10 stone aggregate of 40 mm nominal size) thick for all walls. Brick masonry / Random rubble stone masonry in cement mortar (1:6) (1 cement: 6 coarse sand) shall be provided for foundations.
- 2. PLINTH BEAM, SUNSHADES, ROOF SLAB & LINTELS:** R.C.C. mix of 1:1½:3 (1 cement: 1½ coarse sand: 3 stone aggregate of 20 mm nominal size) for plinth beam over R.R. Masonry / brick work. Necessary sunshades to be provided for all external doors and Windows. Roof slab is to be provided with minimum 100 mm thick R.C.C. mix of 1:1½:3 (1 cement: 1½ coarse sand: 3 stone aggregate of 20 mm nominal size), lintel over Doors & Windows to be provided.
- 3. SUPER STRUCTURE:** Brick masonry in cement mortar (1:6) for 0.23m thick shall be provided.
- 4. FINISHINGS:** 12 mm thick cement plaster with cement mortar (1:6) on finished side of brick work and 15mm thick cement plaster with cement mortar (1:6) on rough side shall be provided.

5. **PAINTING:** Oil bound distemper two coats over one coat of primer for all internal surface walls and snowcem two coats over one coat of primer on external walls.
6. **FLOORING:** Sand filling under floor to be provided. 40 mm thick flooring in C.C. (1:2:4) with neat cement punning and glass strip panels shall be provided.
7. **DOORS & WINDOWS:** For ancillary structures, wooden paneled door, flush door of minimum size 0.90m x 1.98m & 0.76m x 1.98m are to be provided. Wooden Windows of size 0.90m x 1.20m paneled with iron grills and glass ventilators shall be provided.
8. **SANITARY:** Vitrous wash basin, Indian type water closet, ventilating shaft, rain-water pipes, stone ware pipes of 100mm, septic tank & dispersion trench are to be provided as required.
9. **ELECTRIFICATION :** Internal & External Electrification to be provided with necessary suitable Tube light fittings and Fans.
- J. **WEIGHBRIDGE & CABIN ROOM::** Electronic Lorry weigh bridge of 40 / 60MT capacity (as per FCI requirement) of reputed make shall be installed. Cabin room of size 3.3m X 4.8m shall be provided.
- K. **SANTRY POST :** A watchman hut of minimum size 1.2m X 2m is to be provided near the main gate including electrification.
- L. **Land Requirement for conventional type storage godowns (IN IDEAL SHAPE):**

1800 M.T Capacity = 1.0 Acres.(approx) (43560 Square Feet)

Note: The plot of land shall be as far as possible rectangular shaped.